Calls to `notify_one(), notify_all(), wait(), wait_until()`, and `wait_until()` on a single `std::condition_variable` instance are serialized. A call to `notify_one()` or `notify_all()` will wake only threads that started waiting *prior* to that call.

Std::Notify_All_At_Thread_Exit Nonmember Function

Wake all of the threads waiting on a `std::condition_variable` when the current thread exits.

Declaration

```
void notify_all_at_thread_exit(
    condition_variable& cv,unique_lock<mutex> lk);
```

Preconditions

`lk.owns_lock()` is `true`, and the lock is owned by the calling thread. `lk.mutex()` shall return the same value as for any of the lock objects supplied to `wait(), wait_for()`, or `wait_until()` on `cv` from concurrently waiting threads.

Effects

Transfers ownership of the lock held by `lk` into internal storage and schedules `cv` to be notified when the calling thread exits. This notification shall be as-if

```
lk.unlock();
cv.notify_all();
```

Throws

`std::system_error` if the effects can’t be achieved.

Note